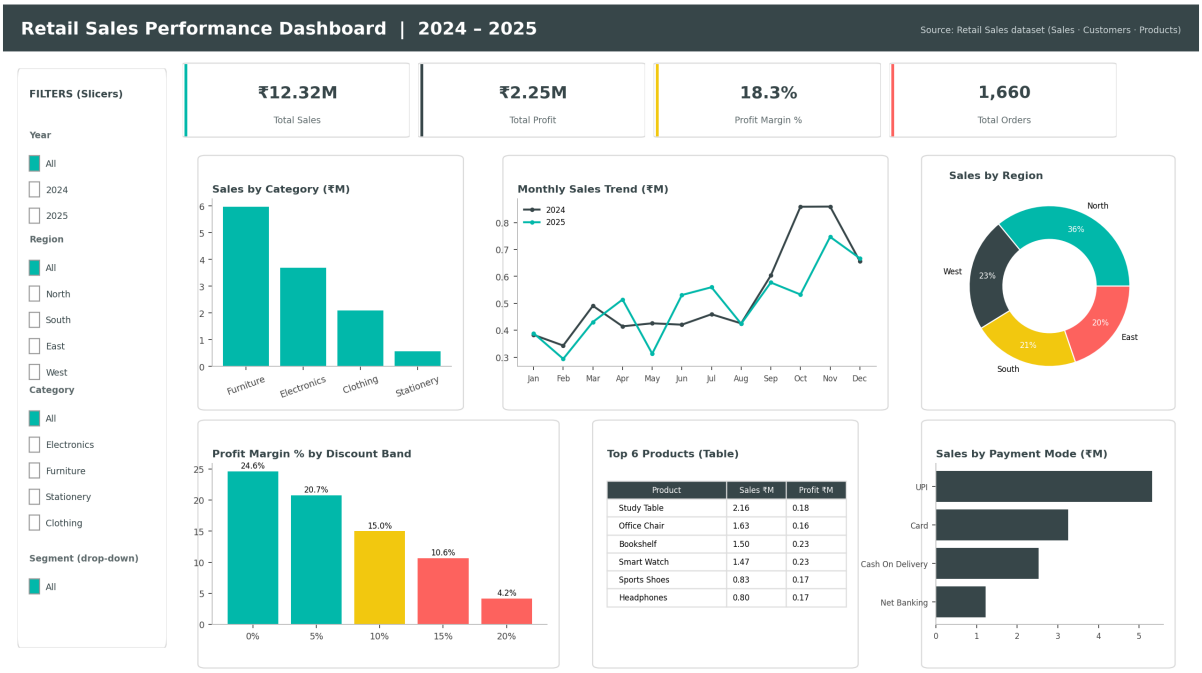
WEEK 5 ASSIGNMENT

Power BI & Business Intelligence

Retail Sales Analysis: Data Preparation, Data Modeling, DAX and Interactive Dashboard



Student Name	[STUDENT NAME]
Roll Number	[ROLL NUMBER]
Course / Program	[COURSE NAME]
Institute	[INSTITUTE NAME]
Instructor	[INSTRUCTOR NAME]
Submission Date	[DATE]
Maximum Marks	100

Contents and Project Overview

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Project overview

This report analyses a **Retail Sales** dataset covering orders placed between 1 January 2024 and 31 December 2025. The data is spread across three related tables: **Sales** (transactions), **Customers** (who bought) and **Products** (what was bought). The dataset is a synthetic practice dataset generated for this assignment (generator script included), designed so that it contains realistic data-quality problems for the cleaning task and enough structure for modeling and DAX.

After cleaning, the model holds **1,660 orders**, **150 customers**, **19 products** in 4 categories and 4 regions. Total sales are **₹12,321,711.45** and total profit **₹2,252,522.48** (margin 18.3%).

Tools used

- Power BI Desktop – data import, Power Query, data model, DAX, report visuals and slicers.
- Source files: `raw_sales.csv, raw_customers.csv, raw_products.csv` (included in the submission).
- Power Query (M) and DAX source code are given in this report and also supplied as separate files.

Task 1 – Power BI Introduction & Data Import (10 Marks)

1.1 Exploring the Power BI Desktop interface

Interface area	What it does
Ribbon (Home, Insert, Modeling, View, Help)	Commands for Get data, Transform data, New measure/column/table, Mark as date table, themes, etc.
Report view	Canvas where visuals, cards, slicers and dashboards are built.
Table (Data) view	Shows the loaded rows and columns of every table so the data and data types can be checked.
Model view	Shows tables as boxes and the relationships between them; relationships are created and edited here.
Fields (Data) pane	Lists all tables, columns and measures that can be dragged onto visuals.
Visualizations pane	Choose the visual type (column, line, donut, matrix, card, slicer…) and assign fields to Axis, Values, Legend, etc.
Filters pane	Applies filters at visual, page or report level.
Power Query Editor	Separate window (Home → Transform data) to clean and shape data before it is loaded.

1.2 Importing the dataset

- Open Power BI Desktop → **Home → Get data → Text/CSV**.
- Select `raw_sales.csv`, review the preview and click **Transform Data** (so the data can be cleaned before loading).
- Repeat for `raw_customers.csv` and `raw_products.csv`. Three queries now appear in the Queries pane: Sales, Customers, Products.

1.3 Tables, columns and data types (after preparation)

Table	Column	Data type	Description
Sales	OrderID	Text	Unique order identifier (primary key)
Sales	OrderDate	Date	Date the order was placed
Sales	CustomerID	Text	Foreign key → Customers
Sales	ProductID	Text	Foreign key → Products
Sales	Quantity	Whole number	Units ordered
Sales	Discount	Decimal number	Discount given (0 – 0.20)
Sales	ShipMode	Text	Standard / Express / Same Day
Sales	PaymentMode	Text	UPI / Card / Cash On Delivery / Net Banking
Customers	CustomerID	Text	Primary key
Customers	CustomerName	Text	Customer full name
Customers	Segment	Text	Consumer / Corporate / Home Office
Customers	Region, City	Text	Geography of the customer
Products	ProductID	Text	Primary key
Products	ProductName, Category	Text	Product and its category
Products	UnitPrice, UnitCost	Fixed decimal (Currency)	Selling price and cost per unit (₹)

1.4 Purpose of Power BI in Data Analytics

Power BI is Microsoft's business-intelligence platform. It connects to many data sources, cleans and models the data, calculates business metrics with DAX and presents the results as interactive reports.

In the data analytics workflow it covers **Get data → Prepare (Power Query) → Model (relationships) → Calculate (DAX) → Visualize → Share**. Its main value is that decision makers can filter and explore the numbers themselves (slicers, cross-filtering, drill-down) instead of reading static spreadsheets, which shortens the time from raw data to a business decision.

Task 2 – Data Cleaning & Preparation (15 Marks)

All cleaning was done in the **Power Query Editor**, so every step is recorded under *Applied Steps* and re-runs automatically when the source file is refreshed. The raw Sales table had **1,730 rows × 10 columns**.

2.1 Data inspection and issues found

Column profiling was switched on (**View → Column quality, Column distribution, Column profile**) and the profile was set to *Entire dataset*. This revealed the following problems:

#	Table	Issue found	Records	Fix applied in Power Query
1	Sales	Duplicate orders (same OrderID repeated)	40	Home → Remove Rows → **Remove Duplicates** on OrderID
2	Sales	Missing Quantity (null)	30	Filter out null/blank rows (the sales amount cannot be computed without Quantity)
3	Sales	Missing ShipMode	25	Replaced null with **"Unknown"** (the order itself is valid and is kept)
4	Sales	Discount stored as text such as "15%" (wrong data type)	15	Strip "%", convert to number and divide by 100; type set to Decimal
5	Sales	Inconsistent text case in PaymentMode (e.g. CARD, NET BANKING)	14	**Transform → Format → Capitalize Each Word**; UPI restored to upper case
6	Sales	OrderDate imported as Text (dd-mm-yyyy)	1,730	**Change Type → Using Locale…** → Date, English (United Kingdom)
7	Sales	Unnecessary columns: *Remarks* (empty) and *Unnamed_Index*	2 cols	Right-click → **Remove Columns**
8	Customers	Customer names in capital letters	37	Format → Capitalize Each Word
9	Customers	Missing Segment	6	Replaced null with "Unknown"

2.2 Transformations performed (Applied Steps – Sales query)

Step	Applied step name	Purpose
1	Source / Promoted Headers	Load CSV, first row becomes column names
2	Removed Columns	Drop Remarks, Unnamed_Index
3	Removed Duplicates	Keep one row per OrderID
4	Filtered Rows	Remove rows with no Quantity
5	Transformed Discount	Text → decimal fraction
6	Replaced ShipMode nulls	Null → Unknown
7	Capitalized PaymentMode	Consistent labels
8	Changed Type (Locale en-GB)	OrderDate → Date, Quantity → Whole number

2.3 Column renaming

The source column names were already compact and without spaces. Descriptive names were applied wherever clarity improved: *Unnamed_Index* was removed, and in the report the calculated fields were named **Sales Amount**, **Cost Amount** and **Profit**. (Right-click header → Rename, or press F2.)

2.4 Before / after result

Measure	Before cleaning	After cleaning
Sales rows	1,730	1,660
Sales columns	10	8
Duplicate OrderIDs	40	0
Null Quantity	30	0
Null ShipMode	25	0 (→ Unknown)
Discount type	Text (mixed)	Decimal number
OrderDate type	Text	Date

2.5 Power Query (M) code for the Sales query

```
let
    Source      = Csv.Document(File.Contents("C:\PowerBI_Week5\data\raw_sales.csv"),
                    [Delimiter=",", Encoding=65001, QuoteStyle=QuoteStyle.Csv]),
    Promoted    = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),

    // 1. remove unnecessary columns
    DropCols    = Table.RemoveColumns(Promoted, {"Remarks", "Unnamed_Index"}),

    // 2. remove the 40 duplicate orders
    NoDuplicates= Table.Distinct(DropCols, {"OrderID"}),

    // 3. drop the 30 rows that have no Quantity
    NoBlankQty  = Table.SelectRows(NoDuplicates,
                    each [Quantity] <> null and [Quantity] <> ""),

    // 4. Discount: text such as "15%" -> number 0.15
    FixDiscount = Table.TransformColumns(NoBlankQty, {{"Discount",
                    each if Text.Contains(_, "%")
                         then Number.FromText(Text.Replace(_, "%", ""), "en-US") / 100
                         else Number.FromText(_, "en-US"), type number}}),

    // 5. missing ShipMode -> "Unknown"
    FixShip     = Table.TransformColumns(FixDiscount, {{"ShipMode",
                    each if _ = null or _ = "" then "Unknown" else _, type text}}),

    // 6. consistent casing in PaymentMode (UPI stays upper case)
    FixPayment  = Table.TransformColumns(FixShip, {{"PaymentMode",
                    each if Text.Upper(_) = "UPI" then "UPI" else Text.Proper(_), type text}}),

    // 7. OrderDate text (dd-mm-yyyy) -> Date, Quantity -> whole number
    Typed       = Table.TransformColumnTypes(FixPayment,
                    {{"OrderDate", type date}, {"Quantity", Int64.Type}}, "en-GB"),
    Typed2      = Table.TransformColumnTypes(Typed,
                    {{"OrderID", type text}, {"CustomerID", type text}, {"ProductID", type text}})
in
    Typed2
```

The M code for the Customers and Products queries is in the file `PowerQuery_M_Code.pq` included with the submission. Click **Close & Apply** to load the cleaned tables into the model.

Task 3 – Data Modeling & Relationships (15 Marks)

3.1 Common columns between tables

Table A	Table B	Common column	Meaning
Sales	Customers	CustomerID	Which customer placed the order
Sales	Products	ProductID	Which product was ordered
Sales	Calendar	OrderDate = Date	Date of the order (time intelligence)

3.2 Relationships created

In **Model view** the columns were dragged between tables (or **Modeling → Manage relationships → New**). The `Calendar` table was created with DAX (Task 4) and marked as a date table.

From (One side)	To (Many side)	Cardinality	Cross-filter direction	Active
Customers[CustomerID]	Sales[CustomerID]	One-to-Many (1:*)	Single (Customers → Sales)	Yes
Products[ProductID]	Sales[ProductID]	One-to-Many (1:*)	Single (Products → Sales)	Yes
Calendar[Date]	Sales[OrderDate]	One-to-Many (1:*)	Single (Calendar → Sales)	Yes

3.3 Data model (star schema)

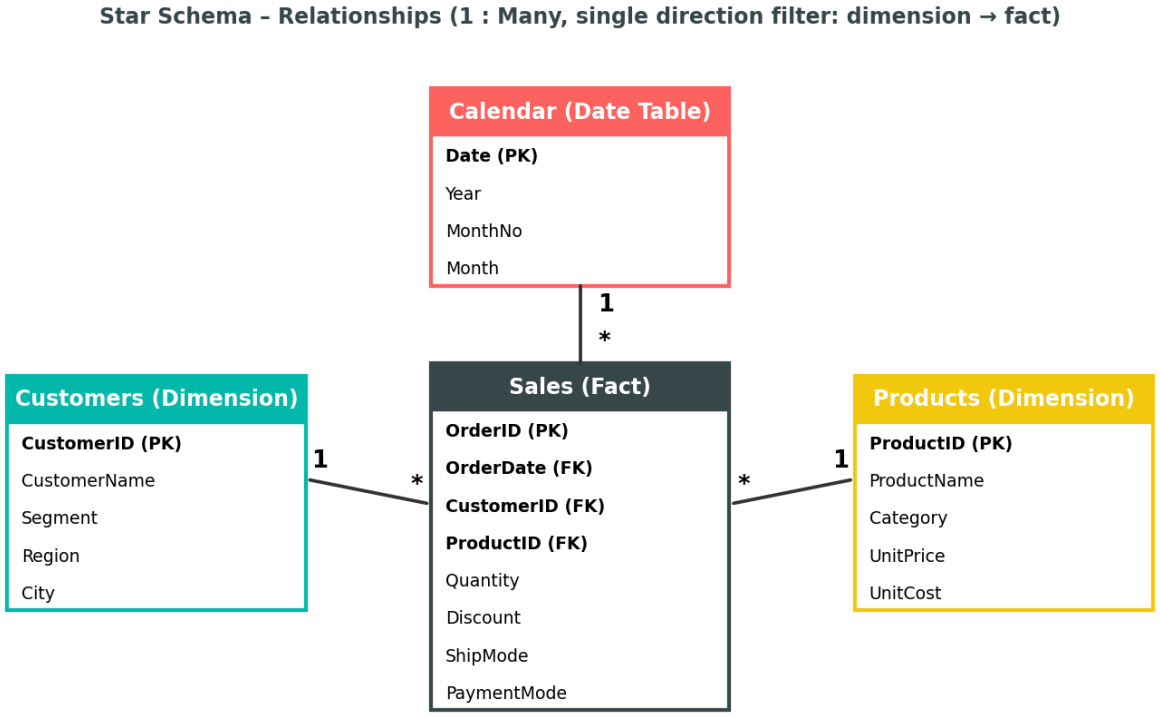


Figure 3.1 – Star schema: one fact table (Sales) surrounded by three dimension tables.

3.4 Why these relationship types were selected

- **One-to-Many:** CustomerID and ProductID are unique in Customers and Products (each person or product appears once) but repeat many times in Sales, because a customer can place many orders and a product can be sold many times. Date is unique in the Calendar table. Power BI detected exactly this pattern.
- **Single direction filtering:** a filter on a dimension (for example Region or Category) flows *down* to the Sales fact table and re-calculates the measures. Single direction avoids ambiguity and keeps the model fast.

- **Star schema:** separating descriptive attributes (dimensions) from transactions (fact) avoids repeating customer and product details on every row, reduces file size and makes slicers behave correctly.
- **No many-to-many or bi-directional relationships** were needed because all keys are unique on the dimension side.

3.5 Validation

A relationship check was done by placing *Products[Category]* and a Sum of Quantity in a table visual: every category showed a different, non-repeating total, confirming that the filter propagates from Products to Sales. No orphan keys existed (every Sales[ProductID] and Sales[CustomerID] was found in its dimension table).

Task 4 – DAX Calculated Columns & Measures (15 Marks)

DAX (Data Analysis Expressions) is the formula language of Power BI. **Calculated columns** are computed row by row when the data is loaded and stored in the table; **measures** are computed on the fly for whatever filters are applied in the report, so they respond to slicers.

4.1 Calendar table and calculated columns

```
Calendar =
ADDCOLUMNS (
    CALENDAR ( DATE ( 2024, 1, 1 ), DATE ( 2025, 12, 31 ) ),
    "Year",    YEAR ( [Date] ),
    "MonthNo", MONTH ( [Date] ),
    "Month",   FORMAT ( [Date], "MMM" )
)

-- Calculated columns in the Sales table
Sales Amount = Sales[Quantity] * RELATED ( Products[UnitPrice] ) * ( 1 - Sales[Discount] )
Cost Amount  = Sales[Quantity] * RELATED ( Products[UnitCost] )
Profit       = Sales[Sales Amount] - Sales[Cost Amount]
Discount Band =
SWITCH ( TRUE (),
    Sales[Discount] = 0,     "0% (No discount)",
    Sales[Discount] <= 0.05, "5%",
    Sales[Discount] <= 0.10, "10%",
    Sales[Discount] <= 0.15, "15%",
    "20%" )
```

RELATED() follows the one-to-many relationship (Sales → Products) to fetch the price and cost of the product on each row.

4.2 Measures and their results

#	Measure	DAX formula	Result (all data)
1	Total Sales	SUM (Sales[Sales Amount])	₹12,321,711.45
2	Total Profit	SUM (Sales[Profit])	₹2,252,522.48
3	Profit Margin %	DIVIDE ([Total Profit], [Total Sales])	18.28%
4	Total Orders (record count)	COUNTROWS (Sales)	1,660
5	Total Units Sold	SUM (Sales[Quantity])	6,696
6	Avg Order Value	AVERAGE (Sales[Sales Amount])	₹7,422.72
7	Max Order Value	MAX (Sales[Sales Amount])	₹47,600.00
8	Min Order Value	MIN (Sales[Sales Amount])	₹144.00
9	Unique Customers	DISTINCTCOUNT (Sales[CustomerID])	150
10	Sales % of Total	DIVIDE ([Total Sales], CALCULATE ([Total Sales], ALLSELECTED ()))	Category share (see 4.3)

#	Measure	DAX formula	Result (all data)
11	Sales LY	CALCULATE ([Total Sales], SAMEPERIODLASTYEAR ('Calendar'[Date]))	2025 → ₹6,341,660.35
12	YoY Growth %	DIVIDE ([Total Sales] - [Sales LY], [Sales LY])	2025: -5.7%

4.3 Sample output – measures sliced by Category

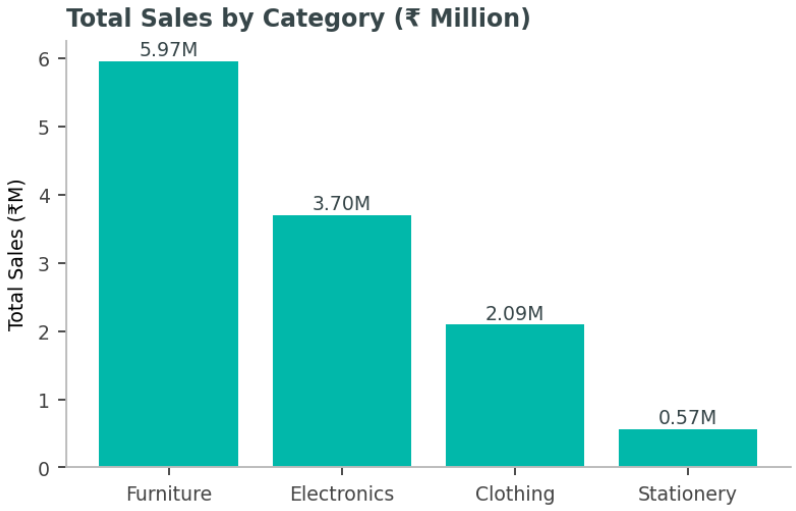
Category	Total Sales	Total Profit	Profit Margin %	Sales % of Total
Furniture	₹5,965,160.00	₹759,660.00	12.7%	48.4%
Electronics	₹3,697,425.00	₹761,166.00	20.6%	30.0%
Clothing	₹2,093,414.45	₹513,374.48	24.5%	17.0%
Stationery	₹565,712.00	₹218,322.00	38.6%	4.6%
Total	**₹12,321,711.45**	**₹2,252,522.48**	**18.3%**	**100.0%**

The results above were validated independently by recomputing the same aggregates from the cleaned data with Python/pandas; the figures match.

Task 5 – Power BI Visualizations (15 Marks)

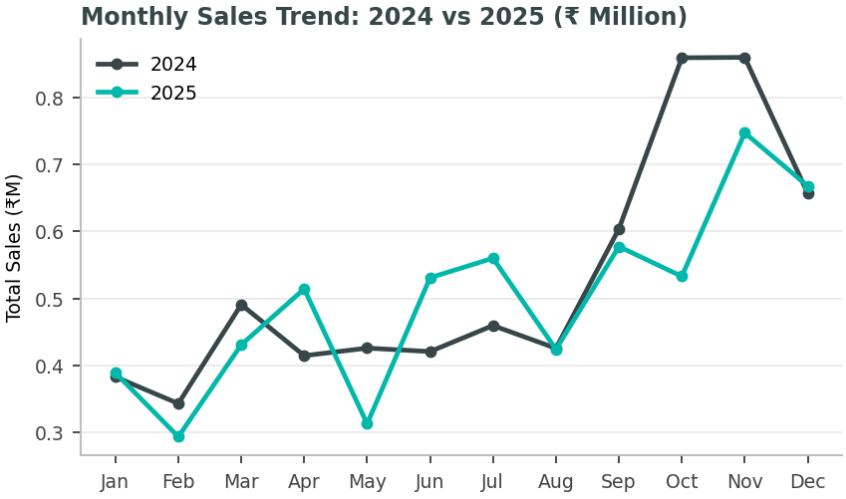
Five core visuals were created on the report canvas. Each uses a meaningful title (Format → General → Title), data labels and axis titles.

5.1 Clustered Column Chart – Sales by Category



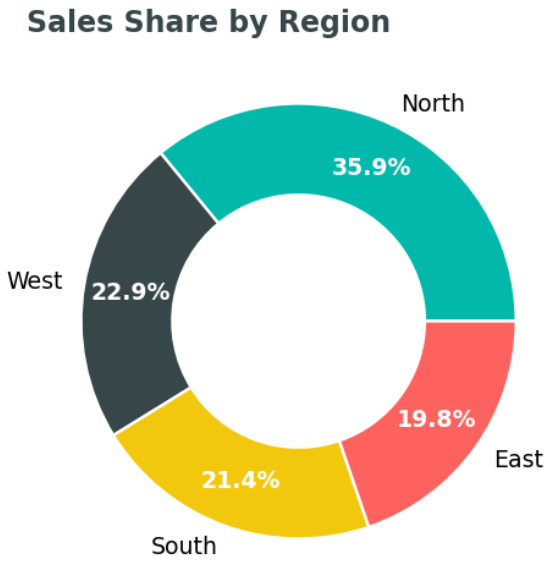
Fields: X-axis = Products[Category]; Y-axis = [Total Sales]. **Why:** compares one measure across a small set of categories. Furniture clearly leads.

5.2 Line Chart – Monthly Sales Trend (2024 vs 2025)



Fields: X-axis = Calendar[Month] (sorted by MonthNo); Y-axis = [Total Sales]; Legend = Calendar[Year]. **Why:** a line shows change over time and the seasonal peak in Oct–Dec is visible in both years.

5.3 Donut Chart – Sales Share by Region



Fields: Legend = Customers[Region]; Values = [Total Sales]. **Why:** shows part-to-whole share across only four regions; North holds the largest slice.

5.4 Matrix – Sales by Category and Region

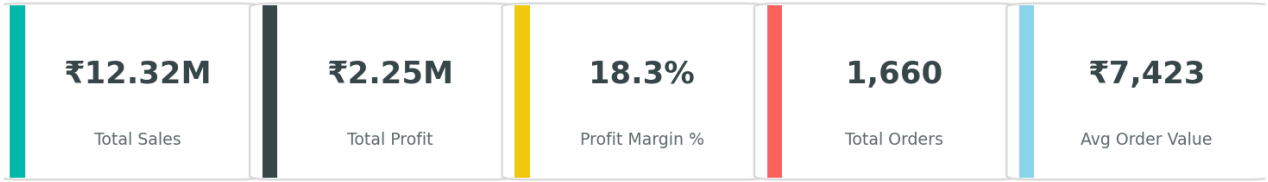
Sales by Category and Region (₹ Million)

	North	South	East	West	Total
Clothing	0.91	0.34	0.49	0.36	2.09
Electronics	1.26	0.65	0.66	1.13	3.70
Furniture	2.07	1.52	1.17	1.21	5.97
Stationery	0.19	0.14	0.12	0.12	0.57
Total	**4.43**	**2.64**	**2.43**	**2.82**	**12.32**

Fields: Rows = Products[Category]; Columns = Customers[Region]; Values = [Total Sales]; row and column grand totals on. **Why:** a cross-tab for exact values and for spotting strong and weak

combinations (e.g. Furniture in North).

5.5 KPI / Card Visuals



Fields: one card per measure – [Total Sales], [Total Profit], [Profit Margin %], [Total Orders], [Avg Order Value]. **Why:** headline numbers are the first thing a manager reads.

Task 6 – Interactive Dashboard & Filters (15 Marks)

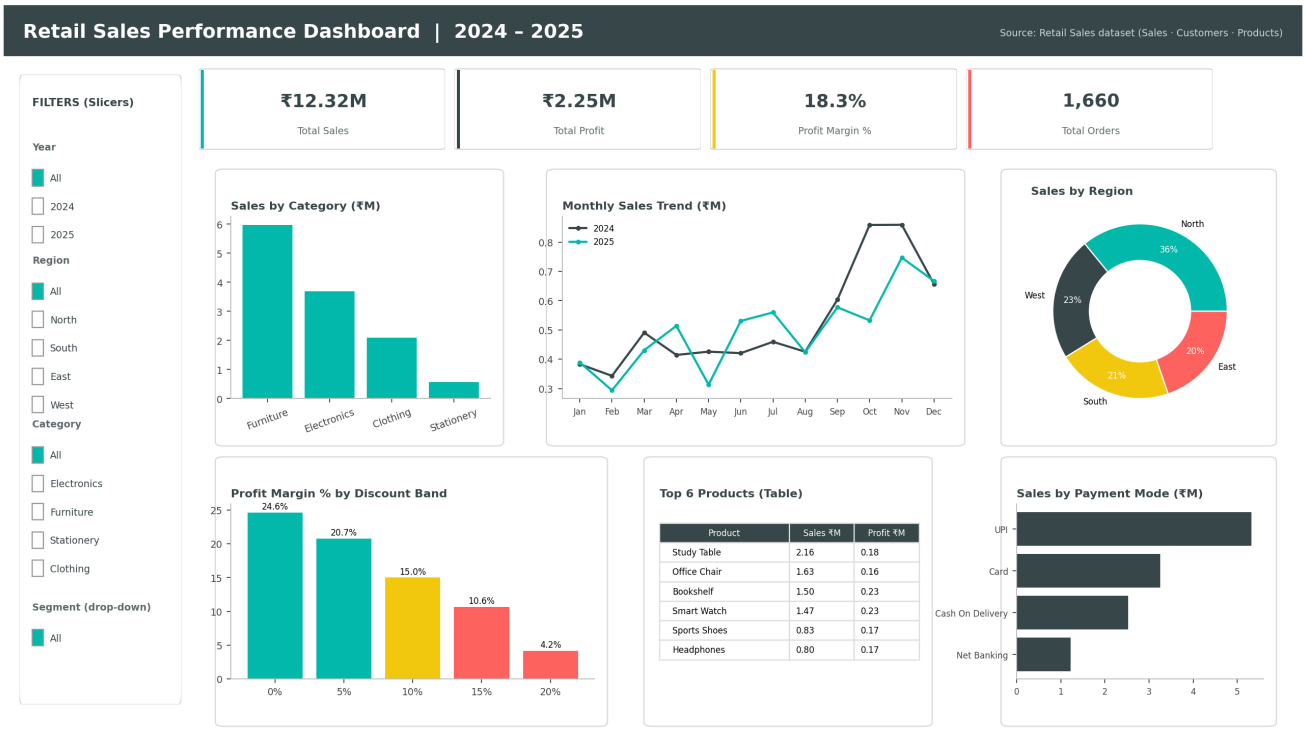


Figure 6.1 – Retail Sales Performance Dashboard (single report page, 16:9 canvas).

6.1 Dashboard components

Requirement	Delivered	Details
≥ 3 KPI cards	4 cards	Total Sales, Total Profit, Profit Margin %, Total Orders
≥ 4 visualizations	6 visuals	Column (Sales by Category), Line (Monthly trend by year), Donut (Region share), Column (Margin % by discount band), Table (Top 6 products), Bar (Sales by payment mode)
≥ 2 slicers	4 slicers	Year (tile buttons), Region (list), Category (list), Segment (drop-down)
Titles and labels	Yes	Every visual has a title, axis titles / data labels and the ₹ unit is stated
Layout	Yes	Title banner → KPI row → trend and distribution visuals → detail row; slicer panel on the left

6.2 How the dashboard is interactive

- **Slicers** filter every visual and KPI card on the page at once (e.g. choose Year = 2025 and Region = North).
- **Cross-filtering / cross-highlighting:** clicking a column (e.g. Furniture) or a donut slice highlights the related data in all other visuals; Ctrl+click selects several items. Controlled with **Format → Edit**

interactions.

- **Tooltips and drill-down:** hovering shows exact values; the Calendar hierarchy lets the line chart drill from Year → Month.
- **Clear filters:** clicking the same selection again (or the eraser icon on a slicer) resets the view.

6.3 Build steps

Step	Action
1	Report view → Format page → Canvas settings: Type = 16:9; Page background = light grey (#F3F2F1).
2	Insert a rectangle (dark #374649) for the title banner and add a text box with the report title.
3	Add 4 **Card** visuals and drag the measures Total Sales, Total Profit, Profit Margin %, Total Orders into Fields.
4	Add Clustered column chart (Category / Total Sales), Line chart (Month / Total Sales / Year legend) and Donut chart (Region / Total Sales).
5	Add Clustered column chart (Discount Band / Profit Margin %), Table (ProductName, Total Sales, Total Profit; Top N filter = 6 by Total Sales) and Bar chart (PaymentMode / Total Sales).
6	Add **Slicer** visuals: Calendar[Year] (Tile), Customers[Region] (List), Products[Category] (List), Customers[Segment] (Dropdown).
7	Align and distribute visuals (Format → Align), apply a consistent colour theme (View → Themes) and set titles. Check interactions with Edit interactions.

Task 7 – Business Insights & Recommendations (15 Marks)

7.1 Key insights from the dashboard

Insight 1 – Overall performance. The business generated **₹12.32M** in sales and **₹2.25M** in profit over 1,660 orders from 150 customers, a profit margin of **18.3%**. The average order is worth ₹7,422.72, ranging from ₹144.00 to ₹47,600.00.

Insight 2 – Furniture drives revenue but not profit. Furniture contributes 48.4% of sales (₹5.97M) but only a 12.7% margin. In contrast, **Stationery** is the most profitable category (38.6% margin) yet gives just 4.6% of sales, which makes it the lowest-selling category. Within products, Study Table is the best seller (₹2.16M) with a thin 8.5% margin, whereas Gel Pen Box sells least (₹55,170.00) but earns 46.3%.

Insight 3 – Strong seasonality. Sales peak in the festive months: **November (₹1.61M)**, October (₹1.39M) and December (₹1.32M). Quarter 4 alone delivers 35.1% of total sales. The weakest month is **February (₹0.64M)**, followed by May (₹0.74M).

Insight 4 – Discounts erode profit. Profit margin falls steadily as the discount rises: 24.6% (no discount), 20.7% (5%), 15.0% (10%), 10.6% (15%) and only **4.2% at 20%**. Orders with 15–20% discount are 26.9% of all orders and 24.8% of sales but only 9.7% of profit (blended margin 7.2%).

Insight 5 – Regional and customer mix. North is the top region (35.9% of sales, ₹4.43M) and **East** the lowest (19.8%, ₹2.43M). Margins are almost the same everywhere (17.7%–18.6%), so regional differences come from sales volume, not pricing. Pune (₹1.31M) and Lucknow (₹1.21M) are the leading cities. The Consumer segment brings 51.7% of sales, while Home Office has the lowest margin (16.6%).

Insight 6 – Payments and year-on-year trend. UPI accounts for 43.2% of sales, followed by Card; Net Banking is smallest (9.9%). Sales in 2025 (₹5.98M) were **5.7% lower** than in 2024 (₹6.34M), so growth has stalled.

7.2 Trends and patterns

- Seasonal pattern: low Jan–May, gradual rise from September, sharp peak October–December in both years.

- Volume vs value: Electronics and Stationery generate many orders (518 and 443) but smaller totals; Furniture has fewer orders (363) with a high order value.
- Inverse relationship between discount level and margin (24.6% → 4.2%).

7.3 Highest and lowest performers

Dimension	Highest (sales)	Lowest (sales)
Category	Furniture (₹5.97M)	Stationery (₹0.57M)
Region	North (₹4.43M)	East (₹2.43M)
Month	November (₹1.61M)	February (₹0.64M)
Product	Study Table (₹2.16M)	Gel Pen Box (₹55,170.00)
Payment mode	UPI (₹5.32M)	Net Banking (₹1.22M)
Category by margin	Stationery (38.6%)	Furniture (12.7%)

7.4 Business recommendations

Recommendation 1 – Put a ceiling on discounts (10%). Orders at 15–20% discount earn a blended margin of 7.2% against 15.0% at a 10% discount. If those orders had earned the 10%-band margin, profit would have been about **₹239,291.05** higher (+10.6%) *(illustrative estimate, assuming the same sales volume)*. Require manager approval above 10% and use bundle offers instead of price cuts for Furniture and Electronics.

Recommendation 2 – Improve the product mix. Furniture makes 48% of sales at a 12.7% margin. Cross-sell high-margin Stationery (38.6%) and Clothing (24.5%) with furniture purchases, and review supplier cost or pricing for Study Table and Office Chair, the two top sellers with the lowest margins.

Recommendation 3 – Prepare for the festive peak and fill the off-season. Secure stock, staff and marketing budget for October–December (over a third of sales), and run targeted campaigns (non-discount offers, loyalty rewards, corporate bulk deals) in February and May to lift the weakest months and reverse the 2025 decline.

Recommendation 4 – Grow East and Home Office. East has the smallest share (19.8%) at a healthy margin, so more marketing and delivery coverage there is likely to add profitable volume. Also fix customer records (3.3% of sales are tagged Segment = Unknown) so segment analysis is reliable.

Appendix – Submission Contents & Reproduction Steps

File / folder	Description
Week5_PowerBI_Assignment.pdf	This report
data/raw_*.csv	Raw Sales, Customers and Products tables (import these into Power BI)
data/clean_*.csv	Cleaned tables (reference output of Task 2)
code/PowerQuery_M_Code.pq	Power Query (M) scripts for the three queries
code/DAX_Measures.dax	Calendar table, calculated columns and measures
code/generate_dataset.py, analyze.py	Python scripts that generate the dataset and validate every figure in this report
screens/	Data model, visuals and dashboard images
README.md	How to rebuild the .pbix file step by step

Reproduction summary

- Load the three raw CSV files → paste the M code into Advanced Editor → Close & Apply.
- Create the Calendar table, relationships, calculated columns and measures from `DAX_Measures.dax`.

- Build the visuals and slicers as described in Tasks 5 and 6, then save the file as `RetailSales_Dashboard.pbix`.

Sample of the cleaned Sales table (first 8 rows after the relationships are applied)

OrderID	OrderDate	CustomerID	ProductID	Qty	Disc.	PaymentMode	Sales Amount (₹)
ORD50001	01-Jan-2024	C1051	P106	2	0.05	Cash On Delivery	2,850.00
ORD50002	01-Jan-2024	C1110	P112	6	0.10	UPI	972.00
ORD50003	02-Jan-2024	C1099	P116	7	0.15	Cash On Delivery	3,564.05
ORD50004	02-Jan-2024	C1095	P110	4	0.00	UPI	7,600.00
ORD50005	03-Jan-2024	C1115	P111	6	0.10	Card	1,296.00
ORD50006	03-Jan-2024	C1117	P108	7	0.20	Cash On Delivery	38,080.00
ORD50007	03-Jan-2024	C1015	P101	3	0.00	UPI	1,950.00
ORD50008	04-Jan-2024	C1029	P105	7	0.15	Cash On Delivery	13,090.00